

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 10: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Theorem 1: tangent and radius

1. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true.

CBSE 2024, 1 mark

SOLUTION

A is true: each tangent is perpendicular to the diameter at its end point, and two lines perpendicular to the same line are parallel.

R is true, but it is not the reason why the tangents are parallel.

Answer: (b)

2 Theorem 2: tangents from an external point

2. AB is a diameter of a circle with centre O . The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90^\circ$. CBSE 2024, 3 marks

SOLUTION

Let PQ touch the circle at C . Join OC .

In $\triangle OAQ$ and $\triangle OCQ$: $OA = OC$ (radii), OQ is common and $QA = QC$ (tangents from Q). So $\triangle OAQ \cong \triangle OCQ$ (SSS), and $\angle AOQ = \angle COQ$.

Similarly $\triangle OBP \cong \triangle OCP$, so $\angle BOP = \angle COP$.

AOB is a straight line: $\angle AOQ + \angle COQ + \angle COP + \angle BOP = 180^\circ$, so $2\angle COQ + 2\angle COP = 180^\circ$.

Hence $\angle POQ = \angle COQ + \angle COP = 90^\circ$.

3. A circle of radius 8 cm with centre O touches the sides AD , DC , CB and BA of a quadrilateral $ABCD$ at P , Q , R and S . $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC . **CBSE 2024, 3 marks**

SOLUTION

Tangents from B are equal: $BR = BS = 24$ cm, so $CR = BC - BR = 30 - 24 = 6$ cm.

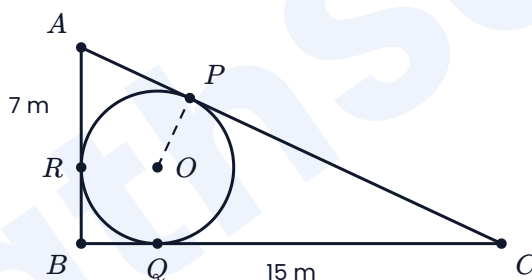
Tangents from C are equal: $CQ = CR = 6$ cm.

In quadrilateral $OPDQ$: $\angle D = 90^\circ$ (given) and $\angle OPD = \angle OQD = 90^\circ$ (radius is perpendicular to tangent), and $OP = OQ = 8$ cm. So $OPDQ$ is a square and $DQ = 8$ cm.

$$DC = DQ + QC = 8 + 6$$

Answer: $DC = 14$ cm

4. A triangular backyard ABC has a right angle at B , with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC , BC and AB at P , Q and R , and $AP = x$ m. (i) Write AR in terms of x . (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x , OR find x and then the radius r of the pit. **CBSE 2024, 4 marks, case study**



SOLUTION

(i) Tangents from A are equal: $AR = AP = x$ m.

(ii) $\angle B = \angle OQB = \angle ORB = 90^\circ$ and $OQ = OR$ (radii), so $BQOR$ is a square.

(iii) $BR = AB - AR = 7 - x$, so $BQ = 7 - x$ and $QC = 15 - (7 - x) = 8 + x$. Tangents from C are equal, so $PC = QC = (8 + x)$ m.

$$AC = \sqrt{7^2 + 15^2} = \sqrt{274} \text{ m, and } AC = AP + PC = 2x + 8. \text{ So } x = \frac{\sqrt{274} - 8}{2} \approx \frac{16.55 - 8}{2} \approx 4.28.$$

$$\text{OR: the radius equals the side of the square, } r = BR = 7 - x = \frac{22 - \sqrt{274}}{2} \approx 2.72 \text{ m.}$$

Answer: (i) x m (ii) a square (iii) $PC = (8 + x)$ m, $x \approx 4.28$; or $r \approx 2.72$ m

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